Politecnico di Milano
Dipartimento di Elettronica, Informazione e Bioingegneria
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Parallel Computing– II part–Thursday, December 21st, 2023

Polimi ID ________ ____________________________

Surname ________ ______________________ **Name** __________________________

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:15m.

Exercise 1 (4 points) ____________________________

Exercise 2 (4 points) ____________________________

Exercise 3 (4 points) ____________________________

Exercise 3 (4 points) ____________________________

Exercise n. 1

Answer the following questions about parallel patterns and briefly explain (without an explanation, the answer will be considered invalid)

A. Please define parallel pattern reduce: assumptions, complexity, and example. (1)

B. Please describe how merge scatter works. (1)

C. Please describe how the Unpack pattern could be implemented. (1)

D. Which loop can be parallelized? (1)

```
for (j=1; j<99; j++)
        for (i=0; i<99; i++)
            a[i][j] = f(a[i][j-1]);
```

Exercise n. 2

Answer the following questions about OpenMP (without an explanation, the answer will be considered invalid).

A. What happens when a variable is declared as private in an OpenMP pragma? (1)

B. It is always necessary to place an explicit barrier at the end of an OpenMP work sharing construct. Is this statement true or false? Explain. (1)

C. Describe what is the effect of the following OpenMP directive: (2)

```
#pragma omp for simd schedule(simd:static, 3)
```

Exercise n. 3

Answer the following questions about CUDA (without an explanation, the answer will be considered invalid).

A. What is the impact of control divergence for the following kernel, assuming blocks of 15x15 threads and an input size of n=1280, m=720? Propose a block size that results in less divergence. (3)

```
__global__ void myKernel(float* d_Pin, float* d_Pout, int n, int m) {
        int Row = blockIdx.y*blockDim.y + threadIdx.y;
        int Col = blockIdx.x*blockDim.x + threadIdx.x;
        if ((Row < m) && (Col < n)) {
            d_Pout[Row*n+Col] = 2*d_Pin[Row*n+Col];
        }
    }
```

B. How does the partitioning of input data impact the performance of a histogram program in CUDA? (1) **This question can be skipped in case of a passed challenge.**

Exercise n. 4

Answer the following questions about parallel programming languages (without an explanation, the answer will be considered invalid).

A. What is a DSL? (2)

B. Compare MPI, Pthreads, OpenMP, and CUDA according to how they describe parallelism and communication (i.e., implicitly/explicitly). (2)